

Sécurisation des accès distants via VPN SSL

BTS SIO Services informatiques aux organisations

Année 2024-2026

Sommaire : Sécurisation des accès distants via VPN SSL

1. PRÉSENTATION DU PROJET (SÉCURITÉ ET MOBILITÉ)
2. CAHIER DES CHARGES (CDC)
3. ANALYSE TECHNIQUE ET ARCHITECTURE
4. MISE EN ŒUVRE TECHNIQUE
 - *Architecture de confiance (PKI)*
 - *Configuration Technique du Serveur*
 - *Filtrage et Sécurité (Firewall)*
 - *Création de l'utilisateur "Test"*
 - *Exportation et Test Client*

1. PRÉSENTATION DU PROJET (SÉCURITÉ ET MOBILITÉ)

L'entreprise OVBS a adopté une politique de travail flexible pour ses collaborateurs et ses administrateurs techniques. Cependant, exposer les services critiques (Active Directory, Serveurs de stock) directement sur Internet présente un risque majeur de cyberattaques.

Le projet consiste à transformer le pare-feu pfSense en une passerelle d'accès distant sécurisée. L'objectif est de créer un "tunnel" chiffré entre le poste nomade de l'administrateur (Mac) et le réseau local de Thiais, garantissant ainsi que l'administration des serveurs reste privée et sécurisée.

2. CAHIER DES CHARGES (CDC)

Besoin Métier :

- Assurer la continuité de l'administration du SI en dehors des locaux d'OVBS.
- Garantir l'intégrité et la confidentialité des échanges de données.
- Supprimer la nécessité d'ouvrir des ports sensibles (RDP, SSH) sur le WAN.

Contraintes Techniques :

- Technologie : Utilisation d'un VPN de type SSL/TLS (OpenVPN) pour sa fiabilité et sa traversée facile des pare-feux.
- Authentification : Mise en place d'une double sécurité par certificat numérique et couple identifiant/mot de passe.
- Routage : Le tunnel doit permettre d'accéder à l'ensemble du LAN (172.16.10.0/24).

Livrables :

- Une autorité de certification (CA) et des certificats valides.
- Un service VPN actif sur le pfSense.
- Un profil client fonctionnel installé sur le poste de l'administrateur.

3. ANALYSE TECHNIQUE ET ARCHITECTURE

Pour répondre aux exigences de cybersécurité d'OVBS, l'architecture repose sur les concepts suivants :

- PKI (Public Key Infrastructure) : Création d'une chaîne de confiance interne. Seuls les clients possédant un certificat signé par la CA d'OVBS peuvent tenter de s'authentifier.
- Chiffrement SSL/TLS : Utilisation de l'algorithme AES-256-GCM, offrant un haut niveau de protection contre l'interception de données.
- Segmentation (Tunnel Network) : Les clients VPN reçoivent une adresse dans un réseau isolé (10.0.8.0/24). pfSense assure ensuite le routage vers le réseau de production.
- Filtrage Périmétrique : * Le port UDP 1194 est ouvert sur le WAN pour accueillir le tunnel.
 - Une interface virtuelle OpenVPN est créée pour contrôler finement les droits d'accès des utilisateurs une fois connectés.

1. Architecture de confiance (PKI)

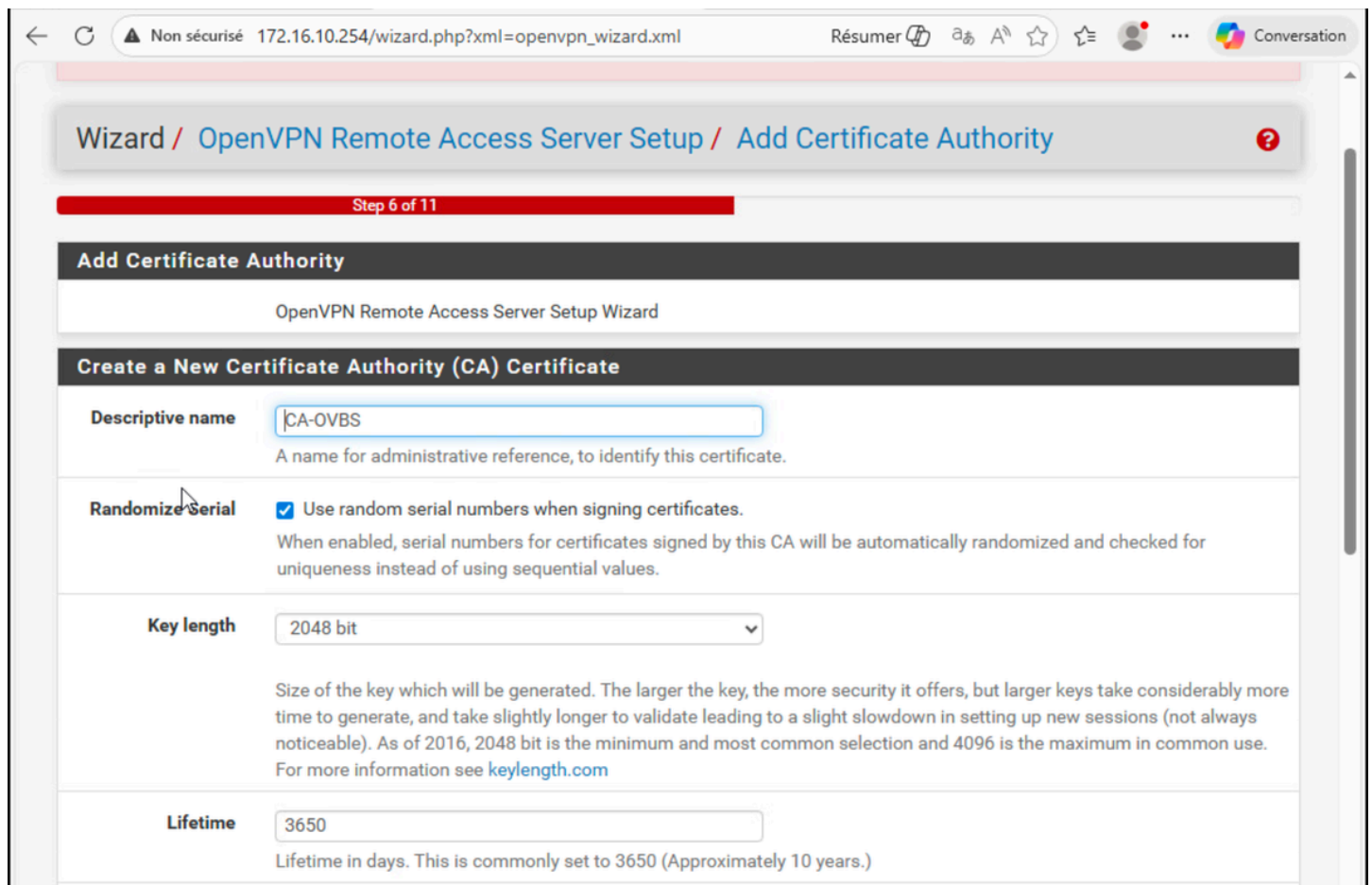
Pour sécuriser le tunnel, j'ai mis en place une infrastructure à clés publiques interne au pfSense :

1. CA (Autorité de Certification) : Création de CA-OVBS pour signer les certificats.
2. Certificat Serveur : Dédié au service VPN pour isoler les secrets cryptographiques.
3. Certificat Client : Généré spécifiquement pour chaque utilisateur autorisé (adminuser).

- Création de l'Autorité de Certification (CA)

C'est elle qui va signer les certificats pour prouver que la connexion est légitime.

1. **Descriptive name** : CA-OVBS.
2. Laisse les autres paramètres par défaut (Key length 2048, etc.).
3. Clique sur **Add new CA**.



The screenshot shows a web browser window with the URL `172.16.10.254/wizard.php?xml=openvpn_wizard.xml`. The page title is "Wizard / OpenVPN Remote Access Server Setup / Add Certificate Authority". A progress bar indicates "Step 6 of 11". The main heading is "Add Certificate Authority". Below this, the text "OpenVPN Remote Access Server Setup Wizard" is displayed. The section "Create a New Certificate Authority (CA) Certificate" contains the following fields:

- Descriptive name**: A text input field containing "CA-OVBS". Below it, a note says: "A name for administrative reference, to identify this certificate."
- Randomize Serial**: A checkbox labeled "Use random serial numbers when signing certificates." which is checked. Below it, a note says: "When enabled, serial numbers for certificates signed by this CA will be automatically randomized and checked for uniqueness instead of using sequential values."
- Key length**: A dropdown menu set to "2048 bit". Below it, a note says: "Size of the key which will be generated. The larger the key, the more security it offers, but larger keys take considerably more time to generate, and take slightly longer to validate leading to a slight slowdown in setting up new sessions (not always noticeable). As of 2016, 2048 bit is the minimum and most common selection and 4096 is the maximum in common use. For more information see keylength.com"
- Lifetime**: A text input field containing "3650". Below it, a note says: "Lifetime in days. This is commonly set to 3650 (Approximately 10 years.)"

Nouvel onglet X pfSense.home.arpa - Wizard: Open X +

Non sécurisé 172.16.10.254/wizard.php?xml=openvpn_wizard.xml Résumer A Conversation

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Wizard / [OpenVPN Remote Access Server Setup](#) / ?

Step:

OpenVPN Remote Access Server Setup

This wizard will provide guidance through an OpenVPN Remote Access Server Setup .

The wizard may be stopped at any time by clicking the logo image at the top of the screen.

Select an Authentication Backend Type

Type of Server Local User Access

NOTE: If unsure, leave this set to "Local User Access."

>> Next

pfSense is developed and maintained by **Netgate**. © ESF 2004 - 2026 [View license](#).

Nouvel onglet X pfSense.home.arpa - Wizard: Open X +

Non sécurisé 172.16.10.254/wizard.php?xml=openvpn_wizard.xml Résumer A Conversation

pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

Wizard / [OpenVPN Remote Access Server Setup](#) / [Server Certificate Selection](#) / ?

Step 7 of 11

Server Certificate Selection

OpenVPN Remote Access Server Setup Wizard

Choose a Server Certificate

Certificate GUI default (69359f0b11be4)

>> Add new Certificate >> Next

pfSense is developed and maintained by **Netgate**. © ESF 2004 - 2026 [View license](#).

- Création du Certificat Serveur

1. Descriptive name : Tape CERT-VPN-OVBS.
2. Clique sur Create new Certificate.

Create a New Server Certificate

Descriptive name

CERT-VPN-OVBS

A name for administrative reference, to identify this certificate.

Key length

2048 bit

Size of the key which will be generated. The larger the key, the more security it offers, but larger keys take considerably more time to generate, and take slightly longer to validate leading to a slight slowdown in setting up new sessions (not always noticeable). As of 2016, 2048 bit is the minimum and most common selection and 4096 is the maximum in common use. For more information see [keylength.com](#)

Lifetime

398

Lifetime in days. Server certificates should not have a lifetime over 398 days or some platforms may consider the certificate invalid.

Common Name

The internal name of the server certificate, used as a part of the certificate subject. Typically set to the hostname of this system. This value is also used as a Subject Alternative Name (SAN). If left blank, the Descriptive Name value will be used for the Common Name and SAN instead.

Country Code

Two-letter ISO country code (e.g. US, AU, CA)

State or Province

Full State of Province name, not abbreviated (e.g. Texas, Indiana, Ontario).

City

2. Configuration Technique du Serveur

On définit les règles du tunnel.

1. Interface : Choisis WAN.
2. Protocol : UDP on IPv4 only (le plus rapide pour un VPN).
3. Local Port : 1194 (le port standard).
4. Cryptographic Settings : Laissez par défaut (AES-256-GCM est excellent).
5. Tunnel Network : C'est le réseau "virtuel" du VPN. 10.0.8.0/24.
6. Local Network : C'est le réseau que l'on veut atteindre. Ici notre LAN

The screenshot displays the 'General OpenVPN Server Information' configuration page. It is divided into three main sections: General OpenVPN Server Information, Endpoint Configuration, and Cryptographic Settings.

General OpenVPN Server Information

- Description:** A text input field containing 'SRV-VPN-OVBS'. Below it, a note states: 'A name for this OpenVPN instance, for administrative reference. It can be set however desired, but is often used to distinguish the purpose of the service (e.g. "Remote Technical Staff"). It is also used by OpenVPN Client Export to identify this VPN on clients.'

Endpoint Configuration

- Protocol:** A dropdown menu set to 'UDP on IPv4 only'. Below it, a note states: 'Protocol to use for OpenVPN connections. If unsure, leave this set to UDP.'
- Interface:** A dropdown menu set to 'WAN'. Below it, a note states: 'The interface where OpenVPN will listen for incoming connections (typically WAN.)'
- Local Port:** A text input field containing '1194'. Below it, a note states: 'Local port upon which OpenVPN will listen for connections. The default port is 1194. This can be left at its default unless a different port needs to be used.'

Cryptographic Settings

- TLS Authentication:** A checkbox labeled 'Enable authentication of TLS packets.' which is checked.
- Generate TLS Key:** A checkbox labeled 'Automatically generate a shared TLS authentication key.' which is checked.
- TLS Shared Key:** A large, empty text input field.

IPv4 Tunnel Network

10.0.8.0/24

This is the virtual network used for private communications between this server and client hosts expressed using CIDR notation (eg. 10.0.8.0/24). The first network address will be assigned to the server virtual interface. The remaining network addresses will be assigned to connecting clients.

Redirect IPv4 Gateway

☐ Force all client generated traffic through the tunnel.

IPv4 Local Network

172.16.10.254

This is the network that will be accessible from the remote endpoint, expressed as a CIDR range. This may be left blank if not adding a route to the local network through this tunnel on the remote machine. This is generally set to the LAN network.

Concurrent Connections

Specify the maximum number of clients allowed to concurrently connect to this server.

Allow Compression

Refuse any non-stub compression (Most secure)

Allow compression to be used with this VPN instance, which is potentially insecure.

Compression

Disable Compression [Omit Preference]

Compress tunnel packets using the chosen option. Can save bandwidth, but is potentially insecure and may expose data. This setting has no effect if compression is not allowed. Adaptive compression will dynamically disable compression for a period of time if OpenVPN detects that the data in the packets is not being compressed efficiently.

Type-of-Service

☐ Set the TOS IP header value of tunnel packets to match the encapsulated packet's TOS value.

Inter-Client Communication

☐ Allow communication between clients connected to this server.

Nouvel onglet

pfSense.home.arpa - VPN: OpenVPN

+

Non sécurisé

172.16.10.254/vpn_openvpn_server.php

Résumer

A

☆

☆

...

Conversation

pfSense

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

VPN / OpenVPN / Servers

?

Servers

Clients

Client Specific Overrides

Wizards

OpenVPN Servers

Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description	Actions
WAN	UDP4 / 1194 (TUN)	10.0.8.0/24	Mode: Remote Access (SSL/TLS + User Auth) Data Ciphers: AES-256-GCM, AES-128-GCM, CHACHA20-POLY1305, AES-256-CBC Digest: SHA256 D-H Params: 2048 bits	SRV-VPN-OVBS	  

+ Add

pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.

3. Filtrage et Sécurité (Firewall)

Le déploiement a nécessité l'ajustement des règles du pare-feu :

1. Firewall Rule : Autorise le trafic entrant sur le port 1194.
2. Règle OpenVPN : Filtrage des accès à l'intérieur du tunnel pour limiter l'utilisateur aux seules ressources nécessaires.

The screenshot shows the 'Firewall Rule Configuration' step of the pfSense OpenVPN Remote Access Server Setup wizard. The breadcrumb trail at the top reads 'Wizard / OpenVPN Remote Access Server Setup / Firewall Rule Configuration'. A red progress bar indicates 'Step 10 of 11'. The main heading is 'Firewall Rule Configuration', followed by the sub-heading 'OpenVPN Remote Access Server Firewall Rules'. The text explains that rules control network traffic and must allow traffic to the OpenVPN server IP and port, as well as traffic from connected clients. It also mentions that automatic rules can be added or configured manually. Below this, there are two sections: 'Traffic from clients to server' and 'Traffic from clients through VPN'. In the first section, a 'Firewall Rule' checkbox is checked, with the description 'Add a rule to permit connections to this OpenVPN server instance from clients anywhere on the Internet.' In the second section, an 'OpenVPN rule' checkbox is checked, with the description 'Add a rule to allow all traffic from connected clients to pass inside the VPN tunnel.' A blue 'Next' button is at the bottom. The footer shows the pfSense logo and text: 'pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.'

Wizard / OpenVPN Remote Access Server Setup / Firewall Rule Configuration

Step 10 of 11

Firewall Rule Configuration

OpenVPN Remote Access Server Firewall Rules

Rules control passing or blocking network traffic as it flows through the firewall.

Rules must be added which allow traffic to reach the OpenVPN server IP address and port, as well as to allow traffic from connected clients inside the OpenVPN tunnel.

The options on this step can add automatic rules to pass this traffic, or rules can be configured manually after completing the wizard.

Traffic from clients to server

Firewall Rule ☒ Add a rule to permit connections to this OpenVPN server instance from clients anywhere on the Internet.

Traffic from clients through VPN

OpenVPN rule ☒ Add a rule to allow all traffic from connected clients to pass inside the VPN tunnel.

[Next](#)

pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.

4. Création de l'utilisateur "Test"

Créer les accès pour les collaborateurs nomades.

1. Action : Aller dans System > User Manager.
2. Création : Utilisateur adminuser avec mot de passe fort.
3. Génération : Cocher la création du certificat utilisateur lié à CA-OVBS.

The screenshot shows a web browser window with the URL `172.16.10.254/system_usermanager.php?act=new`. The page title is "System / User Manager / Users / Edit". The breadcrumb navigation is "System / User Manager / Users / Edit". The main navigation bar includes "Users", "Groups", "Settings", and "Authentication Servers". The "Users" tab is selected.

The "User Properties" section contains the following fields:

- Defined by:** USER
- Disabled:** ☐ This user cannot login
- Username:** adminuser
- Password:** [masked] **Confirm Password:** [empty]
- Full name:** [empty] User's full name, for administrative information only
- Expiration date:** [empty] Leave blank if the account shouldn't expire, otherwise enter the expiration date as MM/DD/YYYY
- Custom Settings:** ☐ Use individual customized GUI options and dashboard layout for this user.
- Group membership:** admins

←

↻

⚠ Non sécurisé 172.16.10.254/system_usermanager.php?act=new

Résumer📄🔍⋮☆🔔👤⋮🌐 Conversation

Create Certificate for User

Descriptive name

CERT-ADMINUSER

Certificate authority

CA-OVBS

Key type

RSA

2048

The length to use when generating a new RSA key, in bits.
The Key Length should not be lower than 2048 or some platforms may consider the certificate invalid.

Digest Algorithm

sha256

The digest method used when the certificate is signed.
The best practice is to use an algorithm stronger than SHA1. Some platforms may consider weaker digest algorithms invalid

Lifetime

3650

Keys

Authorized SSH Keys

Enter authorized SSH keys for this user

IPsec Pre-Shared

✓

Nouvel onglet

✕

pfSense.home.arpa - System: User

✕

+

—

🗖

✕

←

↻

⚠ Non sécurisé 172.16.10.254/system_usermanager.php

Résumer📄🔍⋮☆🔔👤⋮🌐 Conversation

pfSense

COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

↗

WARNING: The 'admin' account password is set to the default value. [Change the password in the User Manager.](#)

System / User Manager / Users

?

Users

Groups

Settings

Authentication Servers

Users

	Username	Full name	Status	Groups	Actions
☐	👁 admin	System Administrator	✓	admins	✎
☐	👤 adminuser		✓		✎🗑

+

 Add

🗑

 Delete

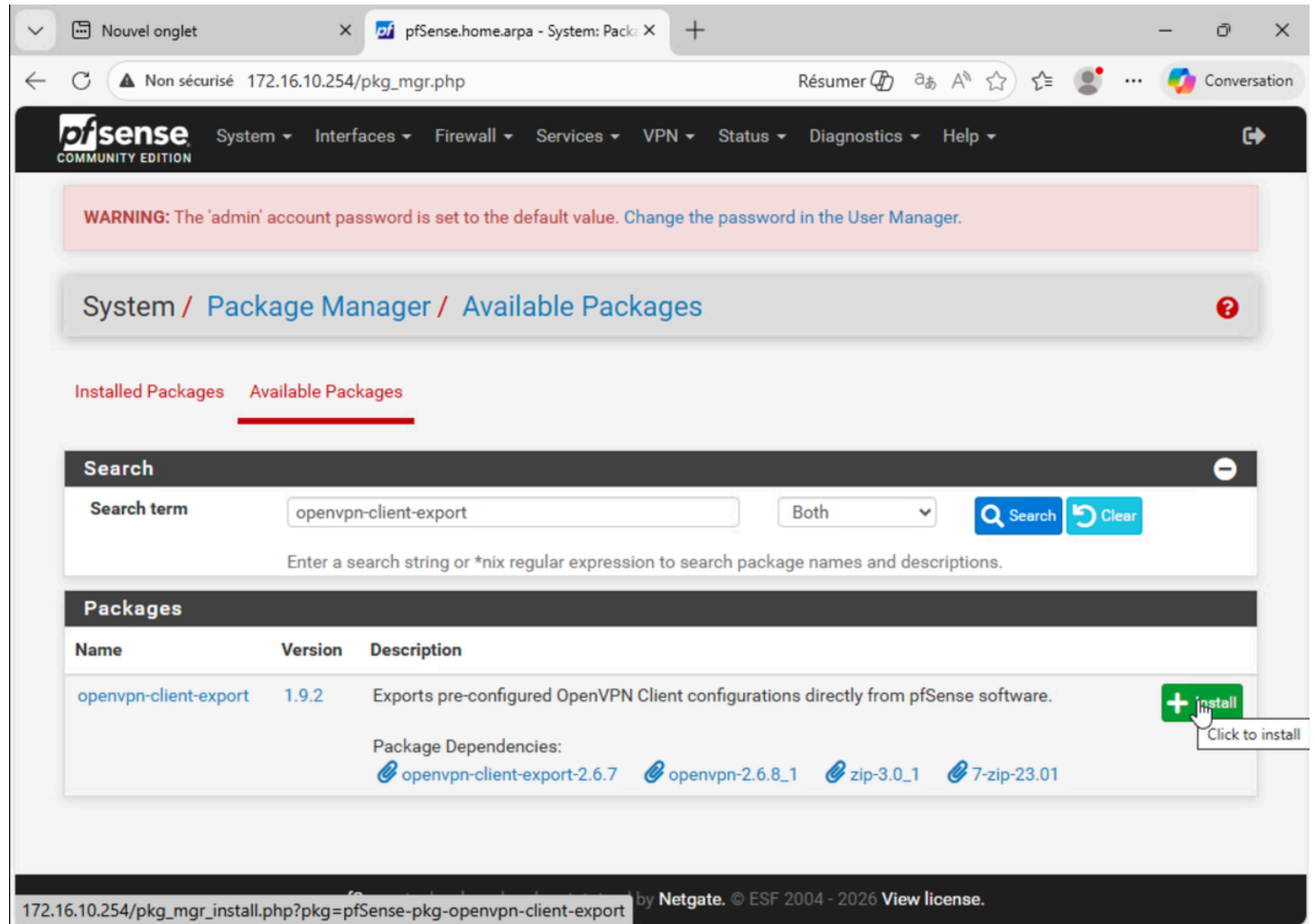
i

pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 [View license.](#)

5. Exportation et Test Client

Le déploiement final.

1. Outil : Installation du package openvpn-client-export.
2. Export : Téléchargement du fichier .ovpn pour l'utilisateur adminuser.
3. Test : Importation dans OpenVPN Connect sur macOS et activation du tunnel.



The screenshot shows the pfSense web interface at 172.16.10.254/pkg_mgr.php. The 'Available Packages' tab is selected. A search for 'openvpn-client-export' yields one result. The package description states it exports pre-configured OpenVPN Client configurations. Below the description, package dependencies are listed: openvpn-client-export-2.6.7, openvpn-2.6.8_1, zip-3.0_1, and 7-zip-23.01. A green '+ install' button is present next to the package name, with a tooltip that says 'Click to install'.

Name	Version	Description
openvpn-client-export	1.9.2	Exports pre-configured OpenVPN Client configurations directly from pfSense software.

Package Dependencies:

- openvpn-client-export-2.6.7
- openvpn-2.6.8_1
- zip-3.0_1
- 7-zip-23.01

← ↻ ⚠ Non sécurisé 172.16.10.254/vpn_openvpn_export.php Résumer ⓘ A ☆ ⌵ 👤 ... Conversation

OpenVPN Clients

User	Certificate Name	Export
adminuser	CERT-ADMINUSER	- Inline Configurations: Most Clients Android OpenVPN Connect (iOS/Android) - Bundled Configurations: Archive Config File Only - Current Windows Installers (2.6.7-lx001): 64-bit 32-bit - Previous Windows Installers (2.5.9-lx601): 64-bit 32-bit - Legacy Windows Installers (2.4.12-lx601): 10/2016/2019 7/8/8.1/2012r2 - Viscosity (Mac OS X and Windows): Viscosity Bundle Viscosity Inline Config

Only OpenVPN-compatible user certificates are shown

If a client is missing from the list it is likely due to a CA mismatch between the OpenVPN server instance and the client certificate, the client certificate does not exist on this firewall, or a user certificate is not associated with a user when local database authentication is enabled.

Clients using OpenSSL 3.0 may not work with older or weaker ciphers and hashes, such as SHA1, including when those were used to sign CA and certificate entries.

javascript:download_begin("confinlineconnect",0, -1)

🍏 Finder Fichier Édition Présentation Aller Fenêtre Aide FR 33 % 🔌

Jaune

Vert

Bleu

Violet

Gris

Bureau

Tous les tags...

Favoris

Applications

Documents

Bureau

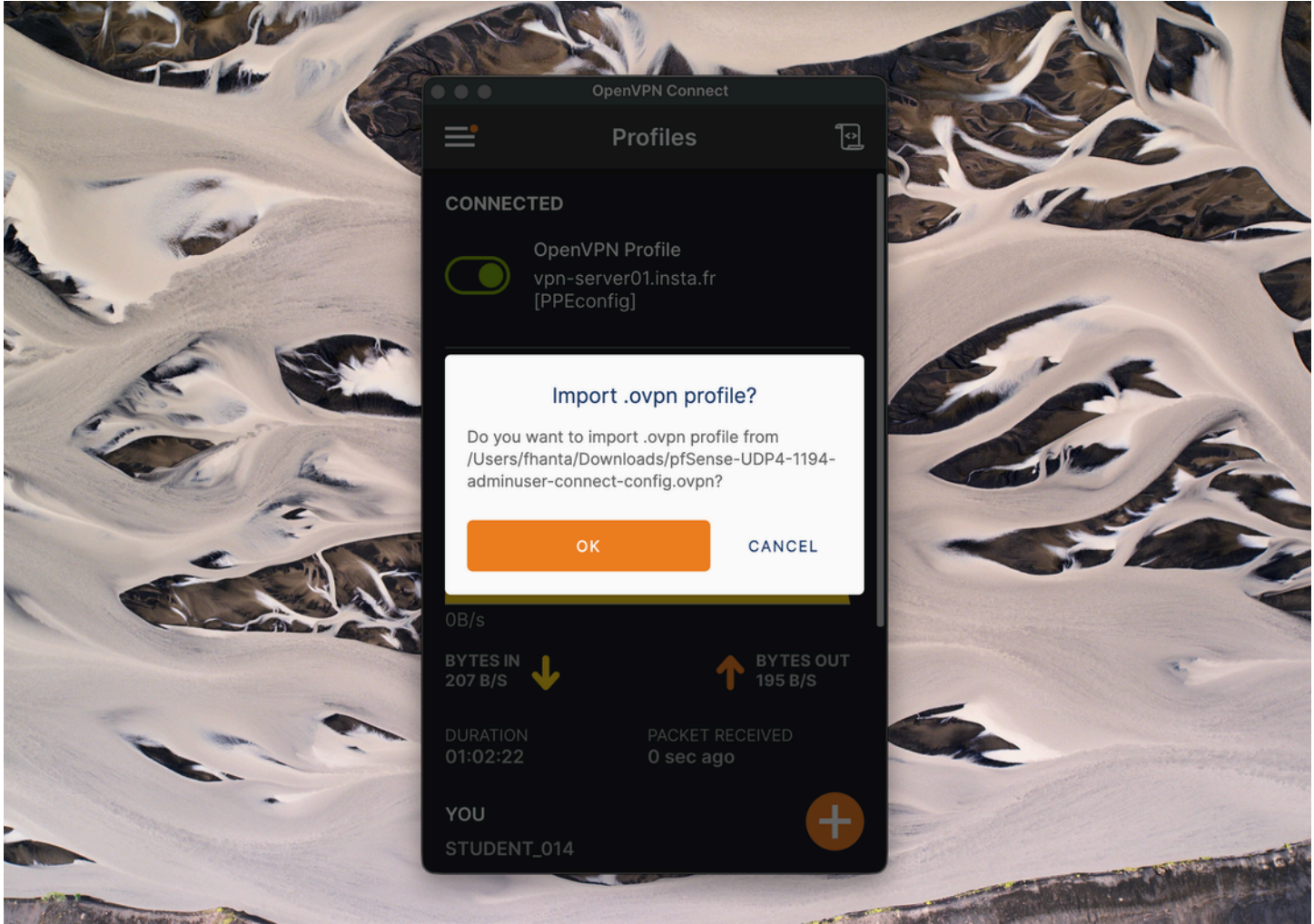
Téléchargements

Capture

open

pfSense-UDP4-1...fig.ovpn

OK CANCEL





Imported Profile

Profile Name

20.0.0.248 [pfSense-UDP4-1194-adminuser-cc

Server Hostname (locked)

20.0.0.248

Username

adminuser

☒ Save password

Password

.....|



PROFILES

CONNECT